Singular Value Decomposition.

Thm. Let $K = \begin{cases} \mathbb{R} \\ \mathbb{C} \end{cases}$, and $A \in M_{m\times n}(K)$. Then $\exists\, P \in \begin{cases} O(m) \\ U(m) \end{cases}$ and $Q \in \begin{cases} O(n) \\ U(n) \end{cases}$ and a "diagonal" matrix

$$D = \left(\begin{array}{ccc|ccc} \lambda_1 & & & & & \\ & \ddots & & & & \\ & & \lambda_k & & & \\ \hline & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{array}\right) \in M_{m\times n}(\mathbb{R}) \text{ with } \lambda_1, \dots, \lambda_k > 0, \text{ where } k = \operatorname{rank} A,$$

s.t. $A = PDQ^{-1}$.

(Recall we defined equivalence of matrices. Here we have orthogonal equivalence.)

Proof. We'll prove only a special case: we assume $\operatorname{rank} A = n$ ($\Leftrightarrow T_A$ is injective. Hence also $n \leq m$). Define $B := A^* \cdot A \in M_{n\times n}(K)$.

We claim that $T_B : K^n \to K^n$ is injective.

Indeed, $\forall\ 0 \neq v \in K^n$ we have

$$\langle T_B v, v\rangle = \langle Bv, v\rangle = \langle A^*Av, v\rangle = \langle Av, Av\rangle > 0. \quad (*)$$

$\Rightarrow T_B v \neq 0. \Rightarrow \operatorname{Ker} T_B = \{0\}$, hence T_B is injective.

Note also that if $\eta \in K$ is an eigenvalue of B, then $\eta \in \mathbb{R}$ (even if $K = \mathbb{C}$) and $\eta > 0$.

Indeed, B is self adjoint ($B^* = (A^*A)^* = A^* \cdot (A^*)^* = A^*A = B$), hence $\eta \in \mathbb{R}$. To see that $\eta > 0$, let v be an eigenvect. of η. From $(*)$ we have: $\langle Bv, v\rangle > 0 \Rightarrow \eta\langle v, v\rangle > 0$ $\Rightarrow \eta > 0$.

By a prev. Thm B is orthog. diag.'ble, hence $\exists\, Q \in \begin{cases} O(n) \\ U(n) \end{cases}$ s.t. $Q^{-1}A^*AQ = \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix}$, where $\eta_j \in \mathbb{R}$, $\eta_j > 0\ \forall j$.

Let $\lambda_j := +\sqrt{\eta_j} > 0$, $j = 1, \dots, n$.

Define $C := AQ \cdot \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \in M_{m \times n}(K)$.

We have $C^* C = \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} \underbrace{Q^{-1} A^* \cdot A Q}_{= \begin{pmatrix} \eta_1 & & 0 \\ & \ddots & \\ 0 & & \eta_n \end{pmatrix}} \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} = I_n$.

It follows that the columns of C form an orthon. system of vectors in K^m. Extend $c_1, \dots, c_n$ to an orthon. basis $c_1, \dots, c_n, f_{n+1}, \dots, f_m$ of K^m and define:

$$\underbrace{\begin{pmatrix} — & \bar{c}_1 & — \\ & \vdots & \\ — & \bar{c}_n & — \end{pmatrix}}_{C^*} \cdot \underbrace{\begin{pmatrix} | & & | \\ c_1 & \cdots & c_n \\ | & & | \end{pmatrix}}_{C}$$

$P := \begin{pmatrix} | & & | & | & & | \\ c_1 & \cdots & c_n & f_{n+1} & \cdots & f_m \\ | & & | & | & & | \end{pmatrix} \in M_{m \times m}(K)$. Clearly $P \in \begin{cases} O(m) \\ U(m) \end{cases}$.

We have: $AQ \begin{pmatrix} \lambda_1^{-1} & & 0 \\ & \ddots & \\ 0 & & \lambda_n^{-1} \end{pmatrix} = C = P \cdot \underbrace{\begin{pmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ \hline & 0 & \end{pmatrix}}_{n} \begin{matrix} \}n \\ \}m-n \end{matrix}$.

$\Rightarrow A = P \cdot \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \\ \hline & 0 & \end{pmatrix} \cdot Q^{-1}$. This proves the Thm. (under the assumption that $\operatorname{rank} A = n$). ∎

For the general case, see the script, chapter 7.5.